WORKSHEET 4.4

Related Rates: Setting Up

Unit 4: Derivatives in Context

Key ideas

- Name the changing quantities, and write the given and wanted rates as derivatives with respect to time, with signs.
- Write **one equation** that relates the quantities at every instant.
- Differentiate both sides with respect to t. Every changing quantity produces its own rate.
- Substitute the values for the instant in question **only after** differentiating.

Worked examples

Example 1 · A sphere

The radius of a sphere changes with time. Relate $\frac{dV}{dt}$ to $\frac{dr}{dt}$.

Solution. $V = \frac{4}{3}\pi r^3$.

$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$.

Example 2 · A fixed hypotenuse

A right triangle has legs x and y and a hypotenuse of fixed length 10. Relate $\frac{dx}{dt}$ and $\frac{dy}{dt}$.

Solution. $x^2 + y^2 = 100$, so $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$.

$x\frac{dx}{dt} + y\frac{dy}{dt} = 0$.

Example 3 · Two changing factors

A triangle has base x and height y, both changing. Find $\frac{dA}{dt}$.

Solution. $A = \frac{1}{2}xy$; use the product rule.

$\frac{dA}{dt} = \frac{1}{2}\left(\frac{dx}{dt}y + x\frac{dy}{dt}\right)$.

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Practice

CORE

1. The radius of a circle is increasing at 3 cm/s. How fast is its circumference increasing?

2. The edge of a cube is increasing at 0.5 cm/s. How fast is its volume increasing when the edge is 4 cm?

3. A point moves along $y = x^2 + 1$ with $\frac{dx}{dt} = 3$. Find $\frac{dy}{dt}$ when $x = 2$.

4. Write as an equation: "the radius is decreasing at 2 centimetres per second".

EXAM LEVEL

5. The area of a square is increasing at 20 cm^2/s. How fast is its side increasing when the side is 5 cm?

6. Two quantities satisfy $xy = 24$. When $x = 4$, x is decreasing at 1 unit per second. Find $\frac{dy}{dt}$.

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7. The volume of a sphere is increasing at 100π cm^3/s. How fast is its radius increasing when $r = 5$ cm?

8. Quantities satisfy $z^2 = x^2 + y^2$ with $z > 0$. When $x = 8$ and $y = 6$, $\frac{dx}{dt} = 1$ and $\frac{dy}{dt} = 3$. Find $\frac{dz}{dt}$.

Question 9 · Free-response practice

The length ℓ of a rectangle is increasing at 4 cm/s and its width w is decreasing at 3 cm/s. At a certain instant $\ell = 12$ cm and $w = 5$ cm.

(a) Find the rate at which the area of the rectangle is changing at that instant.

(b) Find the rate at which the perimeter is changing.

(c) Find the rate at which the length of the diagonal is changing at that instant.

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Worked solutions

1. $\dfrac{dC}{dt} = 2\pi\dfrac{dr}{dt} = 6\pi$ cm/s.

2. $\dfrac{dV}{dt} = 3s^2\dfrac{ds}{dt} = 3(16)(0.5) = 24$ cm^3/s.

3. $\dfrac{dy}{dt} = 2x\dfrac{dx}{dt} = 12$.

4. $\dfrac{dr}{dt} = -2$.

5. $\dfrac{dA}{dt} = 2s\dfrac{ds}{dt}$: $20 = 10\dfrac{ds}{dt}$, so 2 cm/s.

6. $y = 6$. $\dfrac{dx}{dt}y + x\dfrac{dy}{dt} = 0$: $-6 + 4\dfrac{dy}{dt} = 0$, so $\dfrac{dy}{dt} = 1.5$.

7. $100\pi = 4\pi(25)\dfrac{dr}{dt}$, so 1 cm/s.

8. $z = 10$. $10\dfrac{dz}{dt} = 8(1) + 6(3) = 26$, so $\dfrac{dz}{dt} = 2.6$.

9. (a) $\dfrac{dA}{dt} = \dfrac{d\ell}{dt}w + \ell\dfrac{dw}{dt} = 4(5) + 12(-3) = -16$ cm^2/s.

   (b) $\dfrac{dP}{dt} = 2\left(\dfrac{d\ell}{dt} + \dfrac{dw}{dt}\right) = 2(4 - 3) = 2$ cm/s.

   (c) $D = 13$. $D\dfrac{dD}{dt} = \ell\dfrac{d\ell}{dt} + w\dfrac{dw}{dt} = 48 - 15 = 33$, so $\dfrac{dD}{dt} = \dfrac{33}{13}$ cm/s.

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